FLEX: Adaptive Experience Replay for Sample-Efficient Flow Policy Gradients in Vision-Language-Action Models

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Abstract: On-policy policy gradients have driven much of the recent progress of reinforcement learning for large language models, but they are rarely applied to vision-language-action (VLA) models: flow-matching action heads have no tractable likelihood, and a robot collects only tens of episodes between updates, too few for an update that uses each batch once and then discards it. We present FLEX, a flow policy optimization method that reuses recent episodes from a replay buffer across updates while keeping each update close to the current policy. Replayed episodes are down-weighted by their age, the critic learns from new and replayed data and scales the policy gradient by how well it predicts new outcomes, and a consensus update moves only the parameters whose gradient agrees with the optimizer's momentum. FLEX trains the action expert of a pretrained VLA directly, without injected exploration noise, a residual policy, a learned Q-function or human corrections. In the low-data regime, FLEX outperforms on-policy, likelihood-based and noise-steering baselines in both simulated and real-world experiments.

Keywords: Flow policy gradient, Vision-language-action models, Flow matching, Sample efficiency

1 Introduction

Vision-language-action (VLA) models learn manipulation from demonstrations, but deployment errors shift their state distribution and compound [1]. Human corrections reduce this drift at the cost of task-specific effort [2]. Reinforcement learning (RL) instead improves the policy from its own outcomes. FPO and FPO++ enable on-policy updates for flow-matching action heads without direct action likelihoods [3, 4], but estimating advantages and updating from only the current batch limits sample efficiency. FPO++ locomotion experiments use on the order of 10^8 environment steps [4], a prohibitive budget for physical manipulation. RECAP improves a flow-matching VLA on difficult physical tasks through accumulated experience and advantage-conditioned training, using hundreds of rollouts per task and, on some tasks, human corrections [5]. Sample-efficient on-policy flow-policy RL on real robots remains an open problem.

We propose FLEX (**F**low Matching **L**earning from **Ex**perience), which combines an on-policy flow update with accumulated experience in a replay buffer to improve sample efficiency. FLEX combines four components: critic replay with policy-loss scaling by critic accuracy, actor replay with staleness weighting, advantage-prioritized sampling, and a consensus update that moves only parameters whose gradient agrees in sign with their momentum. We evaluate the combined method and its components on simulated manipulation benchmarks, and compare FLEX with baseline methods on physical robotic manipulation tasks.

10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

2 Related Work

RL updates for discretized-action VLAs use PPO or GRPO [6, 7]. Diffusion and flow-matching heads lack direct action likelihoods: FPO and FPO++ use flow-loss surrogates [3, 4], while DPPO, ReinFlow and Flow-SDE use stochastic denoising transitions with tractable ratios [8, 9, 10]. Flow-policy RL reaches simulated VLA manipulation [11], but FPO++ trains locomotion with over 10^8 environment steps [4]. Noise-space RL freezes the VLA and adapts its initial noise, restricting adaptation to supported behavior [12]. FLEX addresses sample efficiency for direct on-policy flow-policy updates on real robots through experience replay.

3 Preliminaries

Flow policy optimization (FPO). State s: observation and instruction; chunks $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_H]$ execute $H_e \leq H$ actions, masked by $m_h = \mathbb{1}[h \leq H_e]$. Advantages $A_k = G_k - V_\phi(s_k)$ use returns G_k, critic V_ϕ. For $t \sim \mathcal{U}[0,1]$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, $\mathbf{A}^t = t\mathbf{A} + (1-t)\boldsymbol{\epsilon}$, $\mathbf{u} = \mathbf{A} - \boldsymbol{\epsilon}$. FPO [3] uses velocity v_θ, loss ℓ_θ, K shared old/current-policy samples:

$$\begin{aligned} \ell_\theta(s, \mathbf{A}; t, \boldsymbol{\epsilon}) &= \|v_\theta(\mathbf{A}^t, t, s) - \mathbf{u}\|_2^2, \\ \widehat{r}_k(\theta) &= \exp\Big[\frac{1}{K}\sum_{j=1}^{K}\big(\ell_{\theta_{\text{old}}}(s_k, \mathbf{A}_k; t_j, \boldsymbol{\epsilon}_j) - \ell_\theta(s_k, \mathbf{A}_k; t_j, \boldsymbol{\epsilon}_j)\big)\Big]. \end{aligned} \tag{1}$$

FPO++. FPO++ [4] uses per-sample ratios,

$$\widehat{r}_{kj}(\theta) = \exp\big(\ell_{\theta_{\text{old}}}(s_k, \mathbf{A}_k; t_j, \boldsymbol{\epsilon}_j) - \ell_\theta(s_k, \mathbf{A}_k; t_j, \boldsymbol{\epsilon}_j)\big). \tag{2}$$

4 FLEX

FLEX retains FPO++'s per-sample flow-loss ratios and pretrained sampler, updating the action expert and critic with a frozen vision-language backbone (Appendix Figs. 4 and 6, Alg. 1, notation in Table 9). Motivating studies (Appendix Figs. 5 and 7) show the failure modes of the on-policy update at this budget that each component addresses, e.g. drift and collapse when a batch is reused for more epochs.

Critic replay and loss scaling. The critic minimizes $\mathbb{E}_{s,G\sim f\mathcal{D}_c+(1-f)\mathcal{R}_c}(V_\phi(s) - G)^2$, mixing a fraction f of the batch $\mathcal{D}_c$ that policy π_{θ_c} collects in iteration c with uniform FIFO replay of M recent episodes (all new while empty). The scoring critic $V_{\bar{\phi}}$ is fitted through batch $c-1$, or first on $\mathcal{D}_0$ at $c = 0$; its fit on $\mathcal{D}_c$ runs while $\mathcal{D}_{c+1}$ is collected (Alg. 1, line 11). Thus for $c > 0$ the new-batch explained variance $E_c = 1 - \mathrm{Var}_{\mathcal{D}_c}[G - V_{\bar{\phi}}(s)]/\mathrm{Var}_{\mathcal{D}_c}[G]$ is measured on data $V_{\bar{\phi}}$ has not fitted; it scales the policy loss by $q_c = \max(q_{\min}, \mathrm{clip}(E_c/E^\star, 0, 1))$, so the policy step shrinks when the critic explains little of the new returns.

Actor replay. From $\mathcal{P}_c = \bigcup_{d=0}^{D}\mathcal{D}_{c-d}$, we draw $|\mathcal{P}_c|$ transitions with replacement per iteration, recompute $A_i = G_i - V_{\bar{\phi}}(s_i)$, and weight age d_i (in iterations) by $w_i^{\text{age}} = 2^{-d_i/h_s}$ with half-life h_s. Replay adds data without the drift of more epochs on one batch: on LIBERO-Spatial, 3 epochs instead of 1 raise the policy change 5-fold and drop success from 63% to 29% (Appendix Fig. 5a,b). Staleness weighting limits old behavior without correcting its distribution and raises mean success from 62.1% to 65.7% (Fig. 5c).

Per-task standardization. For each task κ, the mean μ_κ and unbiased standard deviation σ_κ use window weights $2^{-d_i/h_\kappa}$. Set $\bar{A}_i = (A_i - \mu_{\kappa(i)})/\sigma_{\kappa(i)}$ if $\sigma_{\kappa(i)} > 0$, otherwise $\bar{A}_i = A_i - \mu_{\kappa(i)}$, then, over a minibatch $\mathcal{B}$, $\widetilde{A}_i = (\bar{A}_i - m_\mathcal{B})/\max(s_\mathcal{B}, \sigma_{\min})$, with mean and standard deviation $m_\mathcal{B}$, $s_\mathcal{B}$. Tasks differ in success rate, so each advantage is measured against its own task's recent outcomes; a half-life of 8 iterations instead of 4 lowers iteration-10 success from 60.0% to 45.5% (Fig. 5e).

Advantage-prioritized sampling. The draws mix uniform and advantage-proportional sampling, $p_i = (1-\alpha)/|\mathcal{P}_c| + \alpha|\bar{A}_i|/\sum_{i'\in\mathcal{P}_c}|\bar{A}_{i'}|$ [13], with $w_i^{\text{is}} = p_i^{-1}/(|\mathcal{B}|^{-1}\sum_{i'\in\mathcal{B}}p_{i'}^{-1})$ and $\omega_i = w_i^{\text{is}} w_i^{\text{age}}$. Minibatch normalization makes the correction approximate. Per iteration, 6–38% of transitions come from tasks whose episodes all succeeded or all failed and carry little outcome contrast (Fig. 5d); advantage-proportional draws raise the mean advantage magnitude of sampled transitions above that of the replay window (Fig. 7b).

Policy surrogate. For transition i and sample j, $\widehat{r}_{ij}$ is the FPO++ ratio of Eq. (2) with two changes. The loss covers only executed actions, $\ell_\theta^{\mathbf{m}} = (\sum_h m_h)^{-1}\sum_{h=1}^{H} m_h \|v_\theta(\mathbf{A}^t, t, s)_h - \mathbf{u}_h\|_2^2$, and $\theta_{\text{old}} = \theta_c$, the policy that collected $\mathcal{D}_c$, for every transition, including replayed ones, so clipping bounds the policy change within an iteration while staleness weighting handles the age of replayed behavior. Clipping uses lower and upper limits ε_- and ε_+. With the weights ω_i and advantages $\widetilde{A}_i$ above,

$$\mathcal{L}_{\text{PG}} = -\frac{\sum_{i\in\mathcal{B}}\sum_{j=1}^{K}\omega_i \min\big(\widehat{r}_{ij}\widetilde{A}_i, \text{clip}(\widehat{r}_{ij}, 1-\varepsilon_-, 1+\varepsilon_+)\widetilde{A}_i\big)}{K\sum_{i\in\mathcal{B}}\omega_i}. \tag{3}$$

Consensus update. Adapting masked optimization [14], the pre-update first moment $\mathbf{m}$ and gradient $\mathbf{g}$ define $\mathbf{c} = \mathbb{1}[\text{sign}(\mathbf{m}) = \text{sign}(\mathbf{g})] \vee \mathbb{1}[\mathbf{m} = 0]$. The AdamW step becomes $\Delta\theta' = (\|\Delta\theta\|_2/\|\mathbf{c}\odot\Delta\theta\|_2)\,\mathbf{c}\odot\Delta\theta$, preserving step length. On MT50, parameters whose momentum agrees with the gradient have 3 times the gradient signal-to-noise ratio of the others (Fig. 5f), and the mask gives a larger first-order decrease of the next minibatch's loss than a random mask of equal size (Fig. 7a).

Objective. The loss $\mathcal{L}(\theta) = q_c\mathcal{L}_{\text{PG}}(\theta) + \beta\,\mathbb{E}_{(s,\mathbf{A})\sim\mathcal{D}_{\text{demo}},\,t,\boldsymbol{\epsilon}}[\ell_\theta(s,\mathbf{A};t,\boldsymbol{\epsilon})]$ anchors to SFT demonstrations by behavior cloning with weight β.

5 Experiments

Setup. We evaluate whether FLEX improves a flow-matching VLA with a few tens of online episodes per task. For every method we report the best evaluation success within the rollout budget, across iterations. On few-shot LIBERO [15] the SFT policy uses only 5% of each suite's demonstrations. Each iteration collects 50 episodes, 5 per task, for at most 10 iterations. On MetaWorld MT50 [16], SFT uses all 2,500 demonstrations, each iteration collects 150 episodes over 50 tasks, 3 per task, and success is reported per difficulty group of Seo et al. [17]. On a dual-arm UR5 we train 3 real-world tasks (Appendix Fig. 3) from 100 demonstrations each, with at most 10 iterations of 5 episodes, so at most 50 online episodes per task, and 20 fixed evaluation episodes. As the base, the simulated experiments use $\pi_{0.5}$ [18]. The real-world experiments include the VLA backbone of Cortex 2.6, an improved version of the Cortex 2.0 VLA [19] trained on more industrial robot data. Within each backbone all methods start from the same SFT checkpoint and use the same episode budget; RECAP, an offline method, uses a larger budget on LIBERO and the same budget on MetaWorld MT50 (hyperparameters in Appendix Tables 4–8).

	LIBERO (few-shot)					MetaWorld MT50				
Method	Spatial	Goal	Object	LIBERO-10	Avg.	Easy	Medium	Hard	Very hard	Avg.
SFT base	61.0	62.4	73.0	36.0	58.1	86.1	38.2	56.7	48.0	57.3
FPO++ [3, 4]	65.4	64.7	78.7	36.6	61.4	84.6	38.2	55.0	48.0	56.5
Flow-SDE [10]	63.1	61.7	75.0	34.8	58.7	**88.9**	42.7	53.3	50.0	58.7
DSRL [12]	59.7	57.8	80.5	35.6	58.4	87.5	44.5	63.3	48.0	60.8
RECAP [5]	57.5	59.2	61.5	26.5	51.2	78.6	35.5	53.3	44.0	52.9
FLEX	**72.5**	**66.4**	**82.0**	**40.2**	**65.3**	85.7	**50.9**	**65.0**	**52.0**	**63.4**

Table 1: **Simulation results.** Best evaluation success rate (%) within the rollout budget on the 4 few-shot LIBERO suites and on MetaWorld MT50.

Policy	Success (%)			Completion time vs. $\pi_{0.5}$ SFT		
	Pants folding	T-shirt folding	Shoe unpacking	Pants folding	T-shirt folding	Shoe unpacking
$\pi_{0.5}$ SFT	35	5	40	0%	0%	0%
$\pi_{0.5}$ DSRL	40	0	85	−3%	0%	−4%
$\pi_{0.5}$ FPO++	35	0	65	0%	0%	−4%
$\pi_{0.5}$ **FLEX**	70	15	90	−20%	−5%	−5%
Cortex 2.6 SFT	60	70	60	−22%	−48%	−17%
Cortex 2.6 FLEX	**95**	**85**	**100**	**−29%**	**−52%**	**−24%**

Table 2: **Physical dual-arm UR5.** Success rate and task completion time relative to $\pi_{0.5}$ SFT.

Simulation. FLEX has the highest success on every few-shot LIBERO suite and raises the 4-suite average by 7.2 points, more than twice FPO++'s gain (Table 1). With 5 episodes per task per iteration, the on-policy methods receive little training signal; on LIBERO-10, the hardest suite, FPO++, Flow-SDE and DSRL stay within 1.2 points of the base while FLEX gains 4.2. MT50 shows the same pattern with 3 episodes per task per iteration: FPO++ does not improve on the base, and FLEX is best on the three harder groups and on average. The difference in improvement across groups could be explained by the different proportions of tasks that produce both successful and failed outcomes. For the base policy, the three rollouts collected per task contain both successes and failures in 17.9% of cases in the Easy group and 66.7% in the Hard group. This view is consistent with the notion of increased "learnability" [20]. See Appendix B for a discussion of why RECAP does not improve in this setting.

Physical robot. With 50 online episodes per task, FLEX doubles $\pi_{0.5}$'s success on folding pants (35% to 70%), triples it on folding a T-shirt (5% to 15%) and raises unpacking a shoe from 40% to 90%, while shortening completion time by 5 to 20% (Table 2). As in the simulated experiments, we see the largest relative gain over the baselines where the base SFT policy has the lowest success rate. For the T-shirt task the baselines end at 0%, showing that in the real world too, the inclusion of a replay buffer is critical for success. Cortex 2.6 with FLEX exceeds $\pi_{0.5}$ with FLEX on every task (95%, 85%, 100%), showing that FLEX is able to leverage a stronger pretrained backbone. Qualitatively, Cortex 2.6 learns complete corrective behavior, such as re-grasping after a failed grasp, whereas $\pi_{0.5}$ mainly suppresses failures without recovering from them.

Ablations. We remove each component of FLEX in turn on few-shot LIBERO-Spatial (Table 3). In addition, we replace the full-return advantage with GAE [21] to test whether bootstrapped value targets help with few episodes per task. The three largest regressions come from removing replay, for the actor (8.2 points) and for the critic (8.5 points), and removing the critic-accuracy loss scaling (8.2 points).

Removed component	Δ (points)
Critic replay	−8.5
Actor replay buffer	−8.2
Critic-accuracy loss scaling	−8.2
Staleness weighting	−7.1
Per-task standardization	−6.0
Behavior-cloning regularizer	−5.7
Advantage-prioritized sampling	−5.5
Consensus update	−3.0
GAE instead of the full return	−5.7

Table 3: **Component ablations.**

6 Conclusion

FLEX trains a pretrained VLA's action expert directly with a flow policy gradient under small rollout budgets, without a residual policy, action Q-function, world-model rollouts or one-step distillation [22]. With few episodes per task, the on-policy update is limited by its data rather than its objective; replay, with bounds on stale actions, critic error, task scaling and noisy gradient components, makes earlier batches usable, and replay is the component whose removal costs the most success. On the robot, pretraining and reinforcement learning compound: the industrially pretrained VLA trained with FLEX learns to recover from its errors, whereas FLEX mainly makes $\pi_{0.5}$'s errors less frequent. A limitation is that FLEX needs nonzero success under sparse rewards, and the frozen backbone cannot correct perception errors.

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A Experimental setup and hyperparameters

Shared protocol. Every method starts from the same SFT base and, except RECAP on LIBERO, collects the same number of episodes per iteration. We report the best held-out evaluation within iterations 1–10. On the physical robot, a success gives reward 1 and a failure or timeout 0, with no per-step penalty.

FLEX. FLEX trains the action expert and a value head on the VLM features (Table 4; notation in Table 9).

FPO++. To obtain a strong baseline, we tune FPO++'s hyperparameters for stability. Figure 1 compares this configuration with FPO++ using the corresponding FLEX hyperparameters, excluding settings specific to FLEX. This shows that FLEX's hyperparameters alone do not improve FPO++. The other components are necessary for its success.

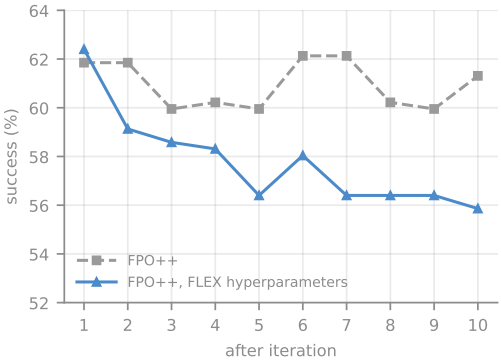


Figure 1: **FPO++ hyperparameter comparison on LIBERO-Spatial.** Performance with the selected FPO++ hyperparameters and the corresponding FLEX hyperparameters.

Flow-SDE. Flow-SDE follows RLinf's $\pi_{0.5}$ PPO configuration [23] except for the episode budget and the batch size (Table 6). It freezes the VLM and trains the action expert and a value head.

DSRL. DSRL follows RLinf's $\pi_{0.5}$ DSRL configuration [23] except for the episode budget and the batch size (Table 7). It freezes the whole VLA and trains a tanh-squashed Gaussian noise policy, whose single noise vector initializes every step of the action chunk; RLinf's 200 gradient updates per 16 episodes set the updates per iteration. On the physical robot, failures and timeouts receive -1 instead of 0.

RECAP. On LIBERO, RECAP collects 3 iterations of 500 episodes, a larger budget than the other methods; on MetaWorld MT50 its 3 iterations of 500 episodes match their budget (Table 8). Each iteration refits the value network from scratch and trains the policy from the SFT checkpoint, on the rollouts of all iterations so far and the demonstrations; the next iteration's episodes are collected with the resulting policy.

[1] RLinf averages the flow-matching loss over the 32 padded action dimensions; with a mean over the 7 real LIBERO dimensions the same update corresponds to 2.2×10^{-6}.

Symbol	Meaning	LIBERO	MT50
N	episodes per iteration	50	150
K	Monte Carlo samples per transition	8	8
ε_-, ε_+	lower and upper clip limits	0.05, 0.15	0.05, 0.15
D	actor replay window (iterations)	3 (Object: 8)	3
h_s	staleness-weighting half-life (iterations)	2	2
α	advantage-proportional share of sampling	0.5	0.5
f, M	share of new transitions in each critic mini-batch, critic buffer (episodes)	0.75, 350	0.75, 350
$E^\star$, $q_{\min}$	explained-variance target, minimum policy-gradient loss scaling	0.5, 0.05	0.5, 0.05
h_κ, $\sigma_{\min}$	task-statistics half-life (iterations), minimum standard deviation	4, 0.15	4, 0.15
β	behavior-cloning regularizer weight	0.01	0.01
–	actor optimizer: AdamW with consensus update (Section 4); learning rate	5×10^{-6} (Object: 10^{-5})	5×10^{-6}
β_1, β_2	AdamW moment decays	0.9 (Goal: 0.5), 0.95	0.5, 0.95
–	actor epochs per iteration, global batch, gradient clip	1, 64, 1.0	1, 1,024, 1.0
–	critic updates per iteration, batch, learning rate, weight decay	128, 64, 10^{-4}, 0.01	128, 64, 10^{-4}, 0.01

Table 4: FLEX hyperparameters.

Hyperparameter	LIBERO	MT50
Episodes per iteration	50	150
Iterations	10	10
Actor learning rate	5×10^{-6}	5×10^{-6}
AdamW betas (β_1, β_2)	0.9, 0.95	0.9, 0.95
Global actor batch	64	1,024
Actor epochs per iteration	1	1
Monte Carlo samples (K)	16	16
Clip limits (ε_-, ε_+)	0.0199, 0.02	0.0199, 0.02
Discount (γ)	0.98	0.99^4
GAE λ	0.99	0.99
Critic learning rate	3×10^{-4}	3×10^{-4}
Critic updates per iteration	64	64
Gradient clip (actor norm)	1.0	1.0

Table 5: FPO++ hyperparameters.

Hyperparameter	LIBERO	MT50
Episodes per iteration	50	150
Iterations	10	10
Actor learning rate	5×10^{-6}	5×10^{-6}
AdamW betas (β_1, β_2), weight decay	0.9, 0.95, 0.01	0.9, 0.95, 0.01
Global actor batch	64	256
Actor epochs per iteration	1	1
PPO clip ε	0.2	0.2
SDE noise level	0.3 (Goal, Object), 0.5 (Spatial, 10)	0.5
Discount (γ)	0.99	0.99^4
GAE λ	0.95	0.95
Value-head learning rate	10^{-5}	10^{-5}
Value-loss clip, Huber δ	0.2, 10	0.2, 10
Gradient clip (norm)	1.0	1.0

Table 6: Flow-SDE hyperparameters.

Hyperparameter	LIBERO	MT50
Episodes per iteration	50	150
Iterations	10	10
Noise dimension	32	32
Noise-policy and Q-network hidden layers	3×128	3×128
Q ensemble size, aggregation	10, mean	10, mean
Discount (γ)	0.999	0.998
Target-network update rate (τ)	0.005	0.005
Initial temperature, target entropy	1.0, -16	1.0, -16
Noise-policy learning rate, Adam betas	10^{-4}, (0.9, 0.96)	10^{-4}, (0.9, 0.96)
Q and temperature learning rate (Adam)	3×10^{-4}	3×10^{-4}
Gradient clip (noise policy / Q and temperature)	3.5 / 10	3.5 / 10
Batch size	128	128
Gradient updates per iteration	625	1,875

Table 7: DSRL hyperparameters.

Hyperparameter	Value
Rollouts	
Episodes per iteration	500
Iterations	3
Collecting policy	previous iteration's, positive suffix
Execution horizon, denoising steps	50, 3
Value network	
Encoder, head	SigLIP2-so400m, Gemma-3-270m
Return bins	201
Reward per step, at failure	-1, -300
Discount factor (γ)	1
Steps, batch size	10k, 256
Learning rate (encoder, head)	5×10^{-5}, 10^{-4}
Advantage labels	
Lookahead (steps)	10
Positive fraction per task	30%
Demonstration chunks	positive
Unconditional dropout	10%
Policy training	
Training chunks	all, 50-step action window
Demonstrations	20, natural share
Trainable parameters	full model
Steps, batch size	10k, 512
Learning rate	10^{-5} (RLinf loss normalization)[1]
Warmup steps, schedule	5k, cosine
Serving	
Prompt, guidance scale	positive suffix, 1

Table 8: RECAP hyperparameters.

Symbol	Meaning
N	episodes per iteration
H, H_e	chunk length, execution horizon
K	Monte Carlo samples per transition
ε_-, ε_+	lower and upper clip limits
D, h_s	actor replay window, staleness-weighting half-life
α	advantage-proportional share of sampling
f, M	share of new transitions in each critic minibatch, critic buffer size
$E^\star$, $q_{\min}$	explained-variance target, minimum policy-gradient loss scaling
h_κ, $\sigma_{\min}$	task-statistics half-life, minimum standard deviation
β	behavior-cloning regularizer weight
β_1	AdamW first-moment decay (momentum memory of the consensus update)

Table 9: Notation of FLEX. Values are listed in Table 4.

B Why RECAP falls below the SFT base

Even given 3x more rollout data than the other methods, RECAP falls below the SFT base on every suite, and we see a steady decline with each cycle. Focusing on LIBERO-Object, we identify two mechanisms driving this: (*i*) RECAP's value network is fitted to the same episodes it subsequently labels leading to near complete memorization of the outcome. This leads to the advantage labels arising mainly from memorization noise. This interpretation is supported by the observation that the advantage indicator I_t correlates weakly with whether the sample comes from a successful episode: 49.8% compared to the base frequency of 52.6%. The authors of RECAP avoid this by co-training the value model on multimodal web data [5]. (*ii*) The base policy is not trained with advantage indicators, and the model shows higher performance when removing the conditioning: conditioning on the advantage indicator consistently results in a decrease of 2.8 ± 2.3 percentage points in success rate across all cycles and suites. Figure 2a shows the MSE to a demonstration, when sampling with eight different noise seeds. We see that this changes no more with the advantage indicator than with the initial noise. Ideally, we would see a significant increase in MSE for the unconditional policy. The attention map in Figure 2b supports this. The action expert attends an order of magnitude more to the images and the task text than to the advantage indicator.

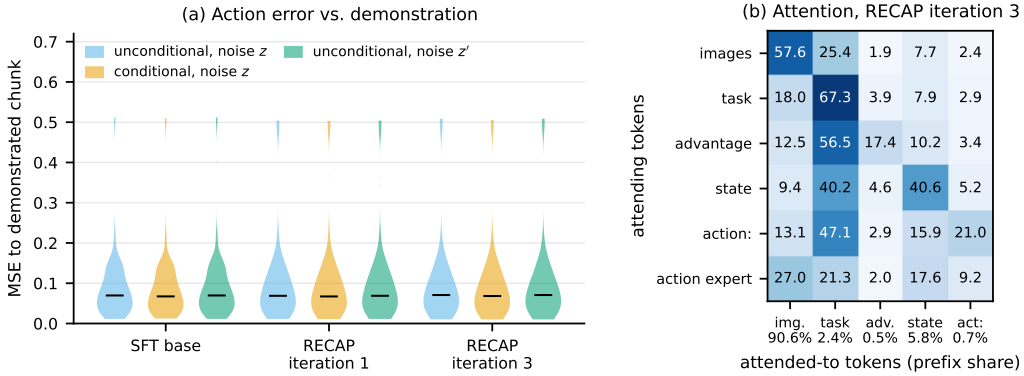


Figure 2: **RECAP advantage conditioning on LIBERO-Object.** (a) MSE between the sampled and the demonstrated action chunk on 50 held-out demonstration states (8 noise draws each), without and with the advantage indicator from the same noise z, and without it from a different noise z'. (b) Share of attention each prefix token group sends to each group after 3 RECAP iterations, mean over heads, layers and 8 held-out states; last row: the action expert at its first denoising step. The indicator changes the sample no more than the noise does and receives 2% of the action expert's attention.

C Supplementary figures

Component-study findings. On LIBERO-Spatial, success rises from 61.85% to 66.21% across 10 iterations with 1 actor epoch but falls from 63.49% to 29.16% with 3; mean approximate KL between successive policies (0.00181 versus 0.00922) indicates greater drift (Fig. 5a,b). Staleness weighting improves mean success over these iterations (65.70% versus 62.10% without it), supporting limited replay of older behavior (Fig. 5c). Per iteration, 6–38% of transitions (mean 22.6%) belong to tasks whose 5 episodes all succeeded or all failed, leaving their advantages without outcome contrast (Fig. 5d). Doubling the task-statistics half-life from 4 to 8 iterations corresponds to lower iteration-10 success (45.50% versus 59.95%), consistent with stale normalization (Fig. 5e). In 3 MetaWorld MT50 runs (iterations 2–4), momentum-agreeing parameters have higher mean gradient signal-to-noise ratio (1.28 versus 0.43) and greater first-order next-minibatch loss decrease per parameter (1.38 versus 0.54 times the step average), supporting consensus filtering (Fig. 5f).

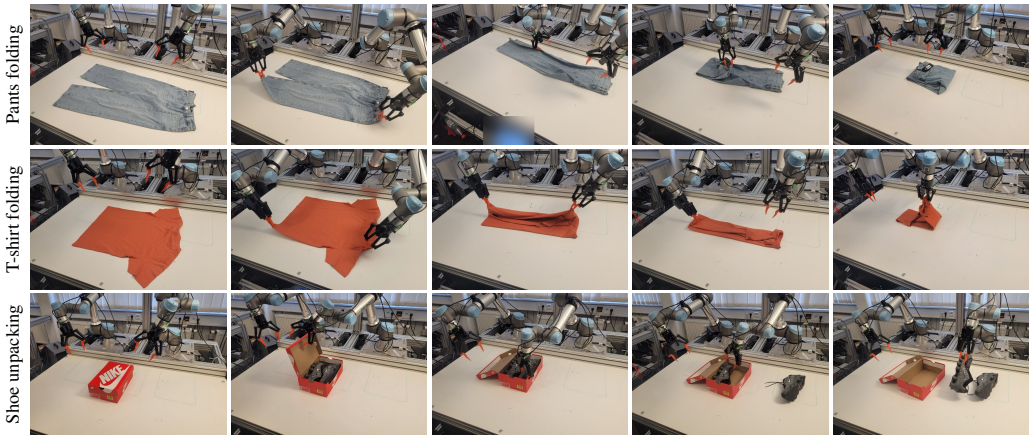


Figure 3: **Physical tasks on the dual-arm UR5.** One episode per row, from the initial state (left) to task completion (right).

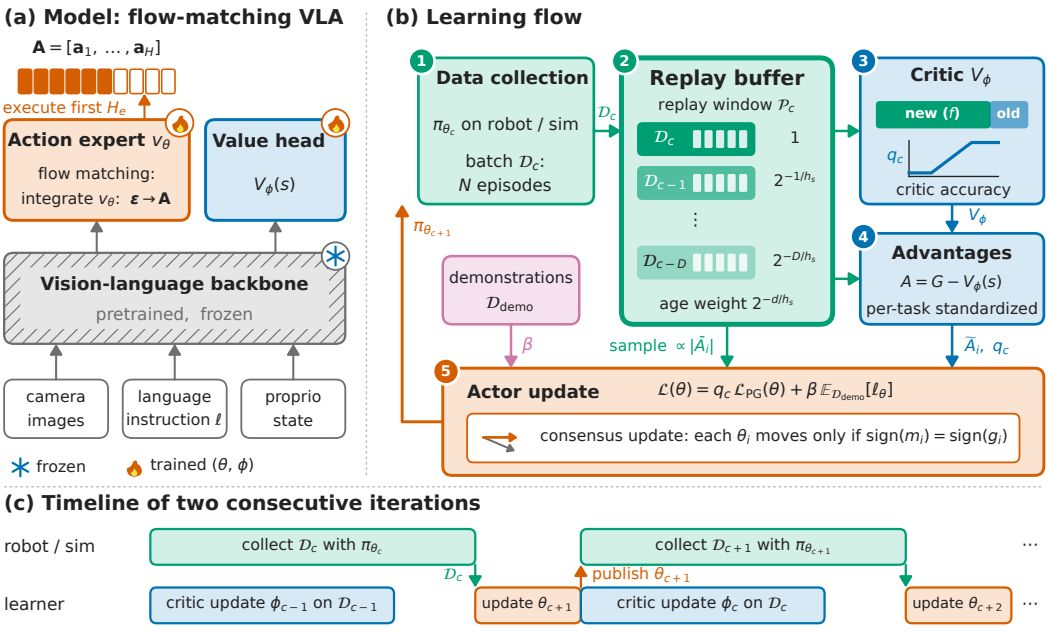


Figure 4: **Overview of FLEX.** (a) Training updates only the action expert v_θ and value head V_ϕ of a flow-matching VLA ($\pi_{0.5}$ in our experiments). (b) Each iteration collects a batch, retains recent batches with staleness weighting, trains the critic on new and replayed episodes, reduces q_c when new-batch accuracy is lower, and updates only the parameters whose momentum and gradient agree in sign (consensus update). (c) The critic update overlaps with collection of the next batch.

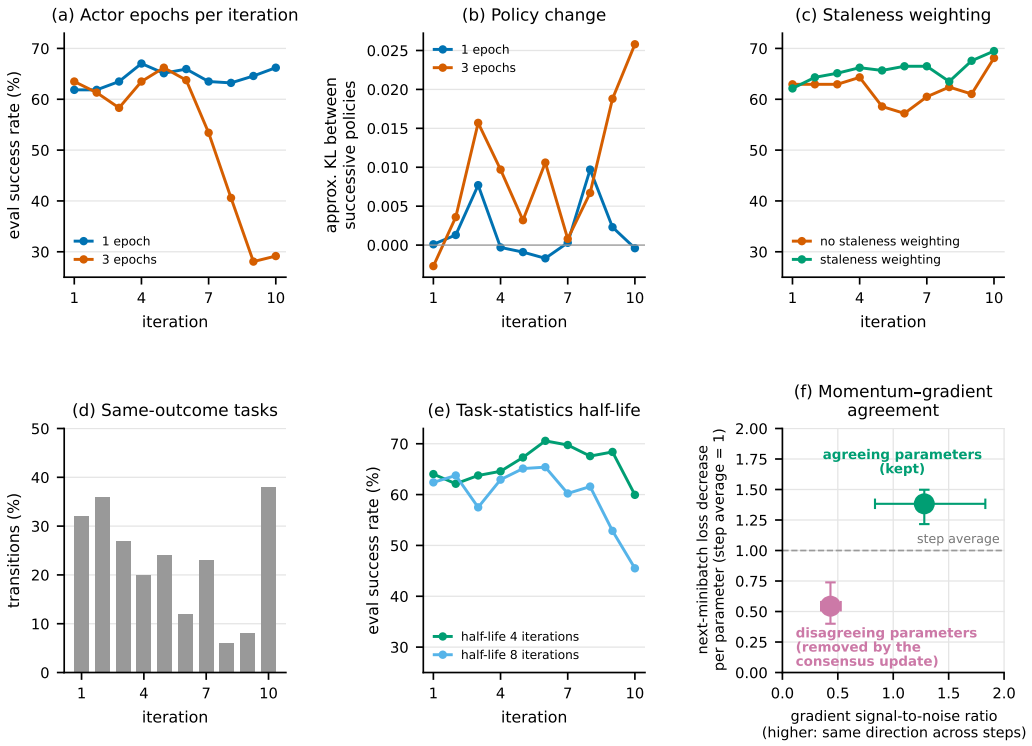


Figure 5: **Studies motivating each component.** (a) Success and (b) policy change with 1 or 3 actor epochs per iteration. (c) Replay with and without staleness weighting. (d) Share of batch transitions from tasks whose episodes all succeeded or all failed. (e) Per-task statistics with half-lives of 4 or 8 iterations. (f) Gradient signal-to-noise ratio and next-minibatch loss decrease per parameter (step average $= 1$) for parameters whose momentum agrees or disagrees in sign with the current gradient, on MetaWorld MT50 (Section 5).

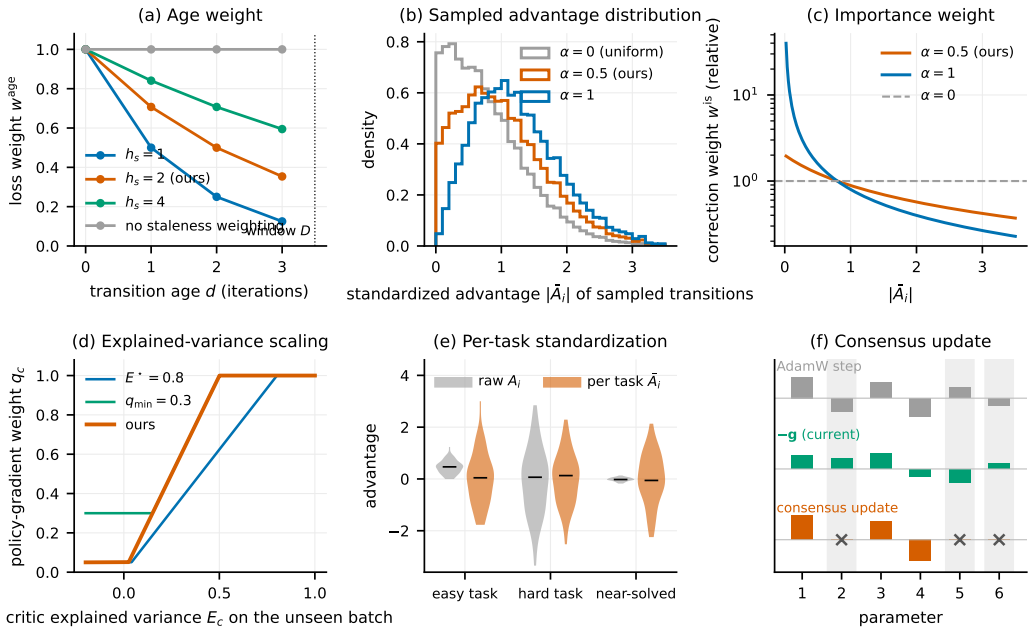


Figure 6: **FLEX weights, sampling and updates.** (a) Replayed transition loss weight versus age. (b) Sampled advantage magnitude in an illustrative replay window. (c) Inverse-probability weight offsetting this sampling preference. (d) Policy-gradient loss scaling by the critic's explained variance on the new batch. (e) Advantages of 3 illustrative tasks before and after per-task standardization. (f) Consensus update on 6 parameters; disagreement between AdamW and $-\mathbf{g}$ (shaded) prevents movement.

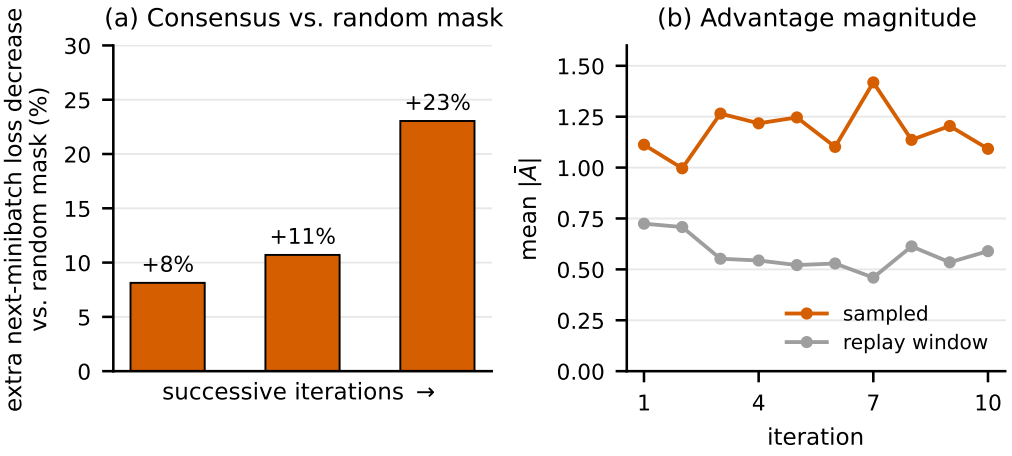


Figure 7: **Consensus update and advantage-prioritized sampling.** (a) Additional first-order decrease of the next minibatch's loss when the consensus mask, instead of a random mask of equal size, selects the parameters of the same AdamW steps (both rescaled to the same step length), on MetaWorld MT50. (b) Mean advantage magnitude of sampled transitions and the replay window on few-shot LIBERO-Spatial.

D Algorithm

Algorithm 1 describes the full procedure of FLEX.

Algorithm 1 FLEX

Require: SFT action expert θ_0 (vision-language backbone frozen), value head V_ϕ with initial parameters ϕ_{-1}, demonstrations $\mathcal{D}_{\text{demo}}$
Require: N episodes per iteration, hyperparameters of Table 9

$\mathcal{R} \leftarrow \emptyset$ ▷ critic buffer
for $c = 0, 1, 2, \dots$ **do**
  Data collection with π_{θ_c}; meanwhile the asynchronous critic update of iteration $c-1$ fits ϕ_{c-1} on $\mathcal{D}_{c-1}$
  collect N episodes: at each decision k, integrate v_{θ_c} from $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$ to a chunk $\mathbf{A}_k$ and execute its first H_e actions
  $\mathcal{D}_c \leftarrow \{(s_k, \mathbf{A}_k, G_k, \kappa_k)\}$ with $G_k = \sum_{k' \geq k} \gamma^{k'-k} r_{k'}$ ▷ κ_k: task of the episode
  Choose the critic that scores this iteration
  if $c = 0$ **then** ▷ pre-V warmup
    $\phi_0 \leftarrow \textsc{CriticFit}(\phi_{-1}, \mathcal{D}_0)$ ▷ pretrain V_ϕ before any advantage is scored
  $\bar{\phi} \leftarrow \phi_{\max(c-1,0)}$ ▷ for $c > 0$: fitted on $\mathcal{D}_0, \dots, \mathcal{D}_{c-1}$, not on $\mathcal{D}_c$
  $E_c \leftarrow 1 - \mathrm{Var}_{\mathcal{D}_c}\big[G_k - V_{\bar{\phi}}(s_k)\big] \big/ \mathrm{Var}_{\mathcal{D}_c}\big[G_k\big]$; $q_c \leftarrow \max\big(q_{\min}, \mathrm{clip}(E_c/E^\star, 0, 1)\big)$
  Policy improvement, scored by $V_{\bar{\phi}}$
  $\theta_{c+1} \leftarrow \textsc{PolicyUpdate}(\theta_c, \mathcal{D}_c, \dots, \mathcal{D}_{c-D}, V_{\bar{\phi}}, q_c)$; publish θ_{c+1} ▷ collection of $\mathcal{D}_{c+1}$ starts
  Asynchronous critic update: runs while $\pi_{\theta_{c+1}}$ collects $\mathcal{D}_{c+1}$
  if $c > 0$ **then**
    $\phi_c \leftarrow \textsc{CriticFit}(\phi_{c-1}, \mathcal{D}_c)$

procedure $\textsc{CriticFit}(\phi, \mathcal{D}_c)$
  for each critic step **do**
    $\mathcal{B}_V \leftarrow$ a fraction f of the transitions sampled from $\mathcal{D}_c$, the rest uniformly from $\mathcal{R}$ (all from $\mathcal{D}_c$ while $\mathcal{R}$ is empty)
    $\phi \leftarrow \phi - \eta_\phi \nabla_\phi \frac{1}{|\mathcal{B}_V|} \sum_{k \in \mathcal{B}_V} \big(V_\phi(s_k) - G_k\big)^2$
  $\mathcal{R} \leftarrow$ the most recent M episodes of $\mathcal{R} \cup \mathcal{D}_c$
  return ϕ

procedure $\textsc{PolicyUpdate}(\theta_c, \mathcal{D}_c, \dots, \mathcal{D}_{c-D}, V_{\bar{\phi}}, q_c)$
  $\mathcal{P} \leftarrow \mathcal{D}_c \cup \dots \cup \mathcal{D}_{c-D}$; a transition i from $\mathcal{D}_{c-d}$ has age $d_i = d$
  $A_i \leftarrow G_i - V_{\bar{\phi}}(s_i)$ for every $i \in \mathcal{P}$ ▷ relabel every transition with the current critic
  $\mu_\kappa, \sigma_\kappa \leftarrow$ mean and unbiased std of $\{A_i\}_{i \in \kappa}$ with weights $2^{-d_i/h_\kappa}$, for every task κ
  $\bar{A}_i \leftarrow (A_i - \mu_{\kappa_i})/\sigma_{\kappa_i}$ ▷ $A_i - \mu_{\kappa_i}$ if $\sigma_{\kappa_i} = 0$
  $w_i^{\text{age}} \leftarrow 2^{-d_i/h_s}$; $p_i \leftarrow \frac{1-\alpha}{|\mathcal{P}|} + \alpha\, |\bar{A}_i| \big/ \sum_{i' \in \mathcal{P}} |\bar{A}_{i'}|$
  $\theta \leftarrow \theta_c$; $\theta_{\text{old}} \leftarrow \theta_c$ ▷ one old policy for every ratio of the iteration
  for $|\mathcal{P}|/B$ steps **do** ▷ as many samples as the window holds, with replacement
    sample B transitions $\mathcal{B} \sim p$ with replacement and a demonstration batch $\mathcal{B}_{\text{demo}}$
    $\widetilde{A}_i \leftarrow (\bar{A}_i - m_{\mathcal{B}})/\max(s_{\mathcal{B}}, \sigma_{\min})$; $w_i^{\text{is}} \leftarrow p_i^{-1} \big/ \big(\frac{1}{B} \sum_{i' \in \mathcal{B}} p_{i'}^{-1}\big)$; $\omega_i \leftarrow w_i^{\text{is}}\, w_i^{\text{age}}$
    $\widehat{r}_{ij} \leftarrow \exp\big(\ell^{\mathbf{m}}_{\theta_{\text{old}}} - \ell^{\mathbf{m}}_{\theta}\big)$ on $(s_i, \mathbf{A}_i; t_j, \boldsymbol{\epsilon}_j)$, $t_j \sim \mathcal{U}[0,1]$, $\boldsymbol{\epsilon}_j \sim \mathcal{N}(0, I)$, $j = 1, \dots, K$
    $\mathcal{L}_{\text{PG}} \leftarrow -\frac{1}{K \sum_i \omega_i} \sum_{i,j} \omega_i \min\big(\widehat{r}_{ij} \widetilde{A}_i,\ \mathrm{clip}(\widehat{r}_{ij}, 1-\varepsilon_-, 1+\varepsilon_+)\, \widetilde{A}_i\big)$; $\mathcal{L}_{\text{demo}} \leftarrow$ mean of ℓ_θ on $\mathcal{B}_{\text{demo}}$
    $\mathbf{g} \leftarrow \nabla_\theta \big[q_c\, \mathcal{L}_{\text{PG}} + \beta\, \mathcal{L}_{\text{demo}}\big]$; $\theta \leftarrow \theta + \Delta\theta'$ ▷ consensus update, Section 4; action expert only
  return θ